

Oakrange Customer Certificate Portal

Product Requirements Document (MVP v1)

Prepared for Oakrange Engineering. This document defines the MVP requirements for a secure customer certificate portal that replaces the current dated WordPress-based workflow.

Core purpose: Oakrange admins upload calibration certificate PDFs, organise them by customer and site, and customers/site managers quickly view and download only the certificate documents they are authorised to access.

1. Product Name

Oakrange Customer Certificate Portal

Alternative internal name: Oakrange Certificate Document Portal

2. Product Summary

Oakrange Engineering needs a modern, secure customer portal where Oakrange admins can upload calibration certificate PDFs, organise them by customer and site, and allow customers or customer site managers to quickly view and download only the certificate documents they are authorised to access.

The portal will replace the current dated WordPress-based workflow with a purpose-built certificate document management system.

The portal is not intended to manage individual calibrated tools in Version 1. Oakrange's uploaded PDF certificates may contain several pages and may cover multiple calibrated tools/items. Therefore, the system will manage certificate PDF documents at customer and site level, not individual tools/assets.

3. Business Objective

The objective is to give Oakrange Engineering a professional, secure and easy-to-use customer portal that improves certificate access for customers while reducing manual admin work for Oakrange staff.

The portal should allow:

- Oakrange admins to upload and manage certificate PDFs efficiently.
- Customers to find and download certificates quickly.
- Site managers to access certificates across multiple assigned sites.
- Oakrange to maintain control over users, permissions and documents.

The portal must improve trust, speed, security and customer experience.

4. Core Product Definition

A secure Oakrange certificate document portal where Oakrange admins upload calibration certificate PDFs, assign them to customers and sites, and customers or site managers can quickly find, view and download only the certificate documents they are authorised to access.

The product is not:

- A general file-sharing platform
- A tool/equipment asset management system
- A customer-managed user administration system
- A full calibration software replacement

5. User Roles

5.1 Oakrange Admin

Oakrange admins are internal Oakrange users. They have full system control.

They can:

- Create and manage customer companies
- Create and manage customer sites
- Create and manage customer users

- Create and manage customer site managers
- Upload certificate documents
- Publish certificates to customer portals
- Replace, void or archive certificates
- Search all customers, sites and certificates
- View audit logs
- Download all certificates
- Manage access permissions

5.2 Customer Site Manager

Customer site managers are users from a customer organisation who may be responsible for one or more sites.

They can:

- Log in securely
- View certificates for assigned sites
- View certificates across multiple assigned sites
- Filter certificates by site
- View certificate details
- View PDFs
- Download PDFs
- See certificate status and dates

They cannot:

- Create users
- Reset passwords for other users
- Upload certificates
- Delete certificates
- Edit certificates
- Manage customer account settings
- Access unassigned sites
- Access other customers' documents

Important rule: Customer site managers only view and download. Oakrange admins create and control users.

5.3 Customer User

Customer users are standard customer portal users.

They can:

- Log in securely
- View certificate documents assigned to their company/site access
- Search and filter their certificates
- View PDFs
- Download PDFs
- See issue/calibration dates, due dates and status

They cannot:

- Upload certificates

- Edit certificates
- Delete certificates
- Create users
- Manage sites
- View audit logs
- Access documents outside their permitted company/site access

6. Role Permissions Matrix

Feature	Oakrange Admin	Site Manager	Customer User
Login	Yes	Yes	Yes
View admin dashboard	Yes	No	No
View customer dashboard	Yes	Yes	Yes
Create customer company	Yes	No	No
Edit customer company	Yes	No	No
Deactivate customer company	Yes	No	No
Create customer sites	Yes	No	No
Edit customer sites	Yes	No	No
Create users	Yes	No	No
Assign users to company/sites	Yes	No	No
Reset user passwords/send invite	Yes	No	No
Deactivate users	Yes	No	No
Upload certificates	Yes	No	No
Publish certificates	Yes	No	No
Replace certificates	Yes	No	No
Void certificates	Yes	No	No
View all certificates	Yes	No	No
View assigned certificates	Yes	Yes	Yes
Download assigned certificates	Yes	Yes	Yes
Search all customers/sites/certificates	Yes	No	No
Search own/assigned certificates	Yes	Yes	Yes
Filter by site	Yes	Yes	Yes, if assigned multiple sites
View audit logs	Yes	No	No
Export certificate list	Yes	Optional MVP	Optional MVP
Manage own password	Yes	Yes	Yes

7. Key Product Principles

7.1 Security First

The system must prevent customers from accessing any certificate that does not belong to them or their assigned sites.

A customer must never be able to access another customer's certificate, even if they guess, copy or manipulate a file URL.

Permissions must be enforced at database/server level, not only by hiding buttons in the user interface.

7.2 UX Is Operational Value

The portal must be easy and fast to use.

Customer UX goal: A customer or site manager should be able to log in, find the right certificate, view it and download it within 30 seconds, without needing instructions from Oakrange.

Admin UX goal: An Oakrange admin should be able to upload and publish a certificate to the correct customer/site in under 60 seconds.

7.3 Admin Control

Oakrange admins must remain in control of customers, sites, users, permissions, certificate uploads, certificate replacements and certificate status.

Customer users and site managers must not be able to manage users or upload documents in MVP.

7.4 Certificate Documents, Not Individual Tools

The portal will manage certificate PDF documents. It will not list individual tools or calibrated items in MVP because the uploaded PDFs already contain the detailed tool/item information.

The system should allow optional tags or notes for search support, but tool-level asset tracking is out of scope for Version 1.

8. Core Workflows

8.1 Admin Login Workflow

Admin visits portal login page

↓

Admin logs in securely

↓

System verifies admin role

↓

Admin is directed to admin dashboard

Admin dashboard should show total customers, total sites, total certificate documents, recently uploaded certificates, certificates expiring soon, expired certificates and recent customer views/downloads.

8.2 Client-First Certificate Upload Workflow

This is the preferred upload workflow because it reduces the risk of uploading to the wrong customer.

Admin logs in

↓

Admin searches for customer by customer ID or company name

↓

Admin opens customer profile

↓

Admin goes to Certificates tab

↓

Admin clicks Upload Certificate

↓

Customer is automatically preselected

↓

Admin selects site if required

↓

Admin selects document type

↓

Admin uploads PDF

↓

Admin enters issue/calibration date

↓

Admin enters due date if relevant

↓

System generates display title and download filename

↓

Admin reviews confirmation

↓

Admin publishes certificate

↓

Certificate becomes visible to authorised customer users/site managers

8.3 Site-Level Certificate Upload Workflow

For multi-site customers, admins should also be able to upload directly from a site record.

Admin opens customer profile

↓

Admin opens Sites tab

↓

Admin selects a site

↓

Admin opens site Certificates tab

↓

Admin clicks Upload Certificate

↓

Customer and site are automatically preselected

↓

Admin uploads PDF and enters certificate dates

↓

System generates display title and download filename

↓

Admin publishes certificate

8.4 Global Certificate Upload Workflow

The portal must also support a global upload workflow for power users or batch-style processing, with clear confirmation before publishing.

Admin logs in

↓

Admin goes to Certificates

↓

Admin clicks Upload Certificate

↓

Admin manually selects customer

↓

Admin selects site if required

↓

Admin selects document type

↓

Admin uploads PDF

↓

Admin enters issue/calibration date

↓

Admin enters due date if relevant

↓

System generates display title and download filename

↓

Admin reviews confirmation

↓

Admin publishes certificate

8.5 Customer Certificate Access Workflow

Customer logs in

↓

Customer lands on dashboard

↓

Customer sees latest certificates first

↓

Customer searches or filters if needed

↓

Customer clicks View PDF or Download PDF

8.6 Multi-Site Manager Workflow

Site manager logs in

↓

Manager sees all assigned sites

↓

Manager can view certificates across all assigned sites

↓

Manager can filter by a specific site

↓

Manager can view/download certificates

The manager should not need separate logins for each site.

9. Admin Requirements

9.1 Admin Dashboard

- Total active customers
- Total sites
- Total certificate documents
- Certificates uploaded recently
- Certificates expiring soon
- Expired certificates
- Recent customer views/downloads
- Quick search
- Quick upload button

9.2 Customer Management

Admins must be able to create a customer company, edit customer details, deactivate/reactivate customers, view customer profiles, view all sites under a customer, view all users under a customer and view all certificates under a customer.

Customer fields should include customer ID, company name, primary contact name, primary contact email, phone number, address, postcode, active/inactive status, created date and notes.

Customer ID should be system-generated and human-readable, such as OAK-1001, OAK-1002 and OAK-1003.

9.3 Site Management

Admins must be able to create customer sites, edit sites, deactivate sites, view site-specific certificates, upload certificates directly to a site and assign users/site managers to sites.

A customer may have one site, multiple sites or no specific site for company-level documents.

9.4 User Management

Admins must be able to create customer users, create site manager users, assign users to a company or selected sites, send invite/reset password, deactivate users and view user access permissions.

Customer site managers and customer users must not create or manage users in MVP.

9.5 Certificate Document Management

Admins must be able to upload certificate PDFs, publish certificates, replace certificates, void certificates, view certificate records, download certificates, filter certificates, search certificates, see original uploaded filename, see customer-facing display name and see upload history.

Certificate document fields should include certificate/document ID, customer/company ID, optional site ID, original uploaded filename, generated display title, generated download filename, document type, issue/calibration date, due date, upload date, uploaded by, status, notes, optional searchable tags and PDF file path/storage reference.

10. Certificate Document Requirements

10.1 Certificate Document Model

The system should treat each uploaded PDF as a certificate document. A certificate document belongs to one customer company, optionally one site, and one uploaded PDF file. A certificate document does not need to belong to an individual tool in MVP.

10.2 Document Types

MVP should support document type as a field. The default type should be Calibration Certificate. The system should allow future document types such as Insurance Certificate, Quality Document, Compliance Document and Other.

10.3 Certificate Status

Recommended stored statuses: Active, Void, Replaced and Archived. Expiry should be calculated from due date. If due_date has passed and status is Active, display as Expired.

Displayed customer-facing statuses should include Active, Expired, Void, Replaced and Archived.

Recommendation for MVP: Active and expired documents are visible. Void documents are hidden from customers by default. Replaced documents can be visible only if admin chooses, but latest document should appear first.

10.4 Historical Certificates

Customers should be able to access historical certificates unless Oakrange voids/removes them. The default view should show newest first, with older certificates lower down.

11. Certificate File Naming Requirements

Oakrange's calibration software generates PDF certificates with random long filenames. The redesigned portal should store the original filename internally and automatically generate a clean customer-facing certificate title and download filename.

11.1 Original Filename

The system must store the original uploaded filename internally for traceability. Example: 837462938472938472.pdf. This should be available to admins but not shown prominently to customers.

11.2 Customer-Facing Display Name

The system should automatically generate a clean customer-facing certificate title and download filename. Admins should not have to manually type a long professional filename.

Recommended display title format: [Site Name] - [Document Type] - [Issue Date]. Example: Sheffield Depot - Calibration Certificate - 12 March 2026.

Recommended download filename format: [Customer Name] - [Site Name] - [Document Type] - [Issue Date].pdf. If no site is selected: [Customer Name] - [Document Type] - [Issue Date].pdf.

11.3 Admin Upload Simplicity

The admin should only need to enter/select customer, site if relevant, document type, issue/calibration date, due date if relevant, PDF file and optional notes/tags. The system should then generate display title, download filename, internal storage filename and status.

Admins may optionally customise the display title, but this should not be required in the normal workflow.

12. Search, Filter and Sorting Requirements

12.1 Admin Global Search

Admins need powerful search. Admin global search should search across customer ID, company name, site name, contact name, email, postcode, certificate title, display filename, original filename, document type, issue/calibration date, due date, upload date, uploaded by and status.

Global search should return grouped results: Customers, Sites, Certificate Documents and Users.

12.2 Admin Customer Search

Admins must be able to find customers by customer ID, company name, contact name, email address, postcode and status.

12.3 Admin Certificate Search and Filters

Admin certificate document page should support search and filters by customer ID, company name, site, document type, issue/calibration date range, due date range, upload date range, uploaded by, status, original filename, display filename and tags/notes.

Upload date is important because admins may remember when they uploaded a certificate.

12.4 Customer Search

Customer search should be simple. Customers/site managers should be able to search by certificate title, site name, document type, issue/calibration date, due date, status and tags where available. Do not overload the customer portal with admin-level filters.

12.5 Sorting Rules

Customer and site manager certificate lists must display newest certificates first by default. Primary sort is issue/calibration date descending. Fallback sort is upload date descending.

ORDER BY issue_date DESC, uploaded_at DESC

Admin certificate management screens should default to recently uploaded first. Admins should still be able to sort by issue/calibration date, due date, customer, site, status, document type and uploaded by.

13. Customer UX Requirements

13.1 Customer Dashboard

Customer dashboard should show a welcome message, latest certificates, number of available certificates, expiring soon count, expired count, quick search and certificate list/table. Latest certificates should appear first.

13.2 Customer Certificate Table

Customer certificate list should show certificate title, site, document type, issue/calibration date, due date, status and actions: View / Download.

Certificate	Site	Issue Date	Due Date	Status	Actions
Sheffield Depot - Calibration Certificate - 12 March 2026	Sheffield Depot	12 Mar 2026	12 Mar 2027	Active	View / Download

13.3 Customer PDF View and Download

Each certificate row should have clear View and Download actions. The download button should not be hidden. The PDF viewer should open the PDF in-browser where possible, with a clear download option always available.

13.4 Customer Confidence Requirements

Each certificate detail page should clearly show customer/company name, site, document type, issue/calibration date, due date, status, certificate title and download button. The customer should not need to open multiple PDFs to guess which one is correct.

14. Site Manager UX Requirements

Site managers may manage multiple customer sites. Their dashboard should show assigned sites, certificate count per site, expiring soon per site, expired per site and latest certificates across all assigned sites.

They should be able to view all certificates across all assigned sites, filter by a specific site, switch between sites without logging out and download certificates.

Your Sites

Sheffield Depot

48 certificates

3 expiring soon

[View certificates](#)

Nottingham Depot

31 certificates

0 expiring soon

[View certificates](#)

Doncaster Depot

22 certificates

1 expired

[View certificates](#)

15. Admin UX Requirements

Admin screens should be fast and practical. Recommended admin navigation: Dashboard, Customers, Sites, Users, Certificates, Expiring Soon, Audit Logs and Settings.

Within a customer profile: Overview, Sites, Users, Certificates, Audit/Notes. Within a site profile: Overview, Users, Certificates, Audit/Notes.

Before publishing a certificate, the system should clearly confirm customer, customer ID, site, document type, issue/calibration date, due date and generated display title. The button label should be Publish Certificate, not just Save.

After publishing, the system should show a clear success message confirming which customer and site the document is now visible to.

16. Security Requirements

16.1 Authentication

The system must use secure authentication. Recommended: Supabase Auth, email/password login, password reset flow, role-based routing and secure sessions. MFA for Oakrange admins can be a future enhancement.

16.2 Authorisation

Authorisation must be enforced server-side and/or at database policy level. Admins can access all companies, sites, users and certificate documents. Site managers can only access assigned company/site certificates. Customer users can only access assigned company/site certificates. Do not rely only on frontend hiding of buttons.

16.3 File Security

Certificate PDFs must be stored in private storage. Files must not be publicly accessible by permanent raw URLs. Recommended approach: private storage bucket, server-side permission check before file access, short-lived signed URLs for viewing/downloading and audit logging when files are viewed/downloaded.

Critical security test: A user must not be able to access another customer's PDF by guessing or copying a file URL.

16.4 Inactive Users

If a user is deactivated, they must not be able to log in, their access to certificates must be blocked, and existing audit logs must remain intact.

16.5 Inactive Customers/Sites

If a customer or site is inactive, admins can still view historic records. Customers should not access inactive customer/site documents unless Oakrange explicitly allows it. MVP should default to restricted access for inactive records.

17. Audit Logging Requirements

The system must record important actions, including user login, certificate uploaded, certificate published, certificate viewed, certificate downloaded, certificate replaced, certificate voided, user created, user deactivated, password reset/invite sent, customer created, site created and permission changed.

Audit log fields should include audit ID, user ID, user role, action, entity type, entity ID, customer ID if relevant, site ID if relevant, timestamp, IP address if available and user agent if available.

Admins should be able to filter audit logs by customer, user, certificate, action and date range.

18. Suggested Database Schema

This is a proposed MVP schema. The exact implementation can be adjusted during development, but the structure should follow this logic.

18.1 profiles

```

id
auth_user_id
full_name
email
role: admin | site_manager | customer_user
company_id, nullable for admins
is_active
created_at
updated_at
last_login_at

```

18.2 companies

```

id

```

customer_id_readable, e.g. OAK-1001

company_name

primary_contact_name

primary_contact_email

phone

address_line_1

address_line_2

town_city

postcode

status: active | inactive

notes

created_at

updated_at

18.3 sites

id

company_id

site_name

site_contact_name

site_contact_email

phone

address_line_1

address_line_2

town_city

postcode

status: active | inactive

notes

created_at

updated_at

18.4 user_site_access

id

user_id

company_id

site_id, nullable if company-wide access

access_type: company | site

created_at

created_by

If access_type = company, user can access all company-level documents and possibly all sites under that company. If access_type = site, user can only access that specific site. This must be implemented carefully.

18.5 certificate_documents

id
company_id
site_id, nullable
document_type
original_file_name
display_title
download_file_name
storage_path
issue_date
due_date
uploaded_at
uploaded_by
published_at
status: active | void | replaced | archived
replaced_by_document_id, nullable
notes
search_tags
created_at
updated_at

18.6 audit_logs

id
user_id
action
entity_type
entity_id
company_id, nullable
site_id, nullable
metadata_json
ip_address
user_agent
created_at

19. Storage Requirements

Recommended private storage structure:

```
/certificates/{company_id}/{site_id_or_company}/{certificate_document_id}.pdf
```

The system should store original filename, generated download filename, internal storage path, file size, MIME type, upload timestamp and uploader ID. Customers should see the generated display/download filename, not the random original filename.

20. Notifications

MVP Recommendation

Email notifications are optional for MVP. The portal can work without automatically notifying customers when a new certificate is uploaded.

Future Enhancement

Later, the system may send new certificate uploaded notifications, certificate expiring soon notifications, certificate expired notifications and password invite emails. For MVP, password reset/invite email is needed, but certificate notifications can be Version 2.

21. Out of Scope for MVP

- Individual tool/asset management
- OCR extraction from PDFs
- Automatic reading of tool data from PDFs
- Customer-side user creation
- Customer-side certificate upload
- Complex reporting dashboards
- Automated expiry email reminders
- QR certificate public verification integration
- Bulk certificate upload
- Engineer mobile upload app
- API integrations with calibration software
- Payment/billing functionality
- White-label customer branding

Some of these may be valuable later, but including them now risks slowing the build and complicating security.

22. Recommended Technology Stack

Recommended stack:

- Frontend: Next.js
- Language: TypeScript
- Styling: Tailwind CSS
- Auth: Supabase Auth
- Database: Supabase Postgres
- Storage: Supabase Storage private bucket
- Permissions: Supabase RLS + server-side checks

- Deployment: Vercel

Why this stack: strong authentication, relational database suited to customers/sites/documents, private file storage, good support for role-based access, good fit for Cursor-based development, maintainable codebase and scalable enough for Oakrange's needs.

23. Suggested Build Phases for Cursor

Do not build everything in one prompt. Use phased implementation.

Phase 1 - Project Setup

Build tasks:

- Create Next.js TypeScript app
- Set up Tailwind
- Set up Supabase client/server utilities
- Set up environment variables
- Create base layout
- Create login page

Acceptance:

- App runs locally
- Supabase connection works
- Login page loads
- Environment variables are documented

Phase 2 - Database Schema and RLS

Build tasks:

- Create tables
- Create roles/profile logic
- Create companies table
- Create sites table
- Create user_site_access table
- Create certificate_documents table
- Create audit_logs table
- Create storage bucket
- Create initial RLS policies

Acceptance:

- Tables created
- Admin can access all records
- Customer cannot access unrelated company data
- Storage bucket is private

Phase 3 - Authentication and Role Routing

Build tasks:

- Implement login/logout
- Create role-based redirects
- Protect admin routes

- Protect customer routes
- Create inactive user checks

Acceptance:

- Admin goes to admin dashboard
- Customer user goes to customer dashboard
- Site manager goes to site manager dashboard
- Inactive users cannot access portal

Phase 4 - Admin Customer/Site/User Management

Build tasks:

- Build admin dashboard
- Build customer list/search
- Build customer create/edit
- Build site create/edit
- Build user create/edit
- Build user assignment to company/sites

Acceptance:

- Admin can create customer
- Admin can create site under customer
- Admin can create user
- Admin can assign user to company/site
- Admin can deactivate user

Phase 5 - Certificate Upload

Build tasks:

- Build client-first upload
- Build site-level upload
- Build global upload
- Implement automatic display name generation
- Store original filename
- Store PDF securely
- Create certificate document record
- Publish certificate

Acceptance:

- Admin can upload from customer profile
- Admin can upload from site profile
- Admin can upload from global certificate page
- Customer/site is preselected where appropriate
- Generated display name is correct
- Original filename is stored
- PDF is private

Phase 6 - Customer and Site Manager Portals

Build tasks:

- Build customer dashboard
- Build certificate list
- Build search/filter/sort
- Build PDF view/download
- Build site manager multi-site dashboard

Acceptance:

- Customer sees only authorised certificates
- Site manager sees only assigned site certificates
- Newest certificates appear first
- View/download works
- Unauthorised PDF access is blocked

Phase 7 - Audit Logs

Build tasks:

- Log upload
- Log publish
- Log view
- Log download
- Log user creation
- Log permission changes
- Build admin audit log view

Acceptance:

- Admin can see who uploaded a certificate
- Admin can see when a customer viewed/downloaded certificate
- Audit logs cannot be edited by customer users

Phase 8 - Polish, Testing and Security Review

Build tasks:

- Improve UI
- Test mobile responsiveness
- Test edge cases
- Test RLS/security
- Test file access
- Test search and filters
- Fix bugs
- Prepare deployment

Acceptance:

- No customer can access another customer's data
- No public PDF leakage
- Customer can download latest certificate in under 30 seconds
- Admin can upload certificate in under 60 seconds
- System works on desktop and mobile

24. MVP Acceptance Criteria

24.1 General Acceptance Criteria

- The portal has three roles: admin, site manager and customer user.
- Oakrange admins control users, customers, sites and certificate documents.
- Customer users and site managers can only view/download authorised documents.
- The portal manages certificate PDF documents, not individual tools.
- PDFs are stored securely and are not publicly accessible.
- The UI is responsive on desktop, tablet and mobile.

24.2 Admin Acceptance Criteria

- An Oakrange admin can log in and access the admin dashboard.
- An admin can create, edit and deactivate customer companies.
- An admin can create, edit and deactivate customer sites.
- An admin can create customer users and site managers.
- An admin can assign users to a whole company or selected sites.
- An admin can search customers by customer ID or company name.
- An admin can search certificate documents by company, site, upload date, issue date, due date, status and document type.
- An admin can upload a certificate from inside a customer profile without manually reselecting the customer.
- An admin can upload a certificate from inside a site profile without manually reselecting the customer or site.
- An admin can upload a certificate from a global upload page by manually selecting the customer and site.
- An admin can upload and publish a certificate in under 60 seconds in normal use.
- The system stores the original random PDF filename internally.
- The system automatically generates a clean customer-facing display title and download filename.
- The system clearly confirms which customer and site the certificate is being published to.
- An admin can replace, void or archive a certificate.
- An admin can view audit logs for uploads, views and downloads.

24.3 Customer Acceptance Criteria

- A customer user can log in securely.
- A customer user can see only certificates they are authorised to access.
- A customer user cannot access another customer's certificates.
- A customer user sees certificates newest first by default.
- A customer user can search/filter their certificate list.
- A customer user can view a certificate PDF.
- A customer user can download a certificate PDF.
- A customer user can find and download the latest certificate in under 30 seconds in normal use.
- The customer-facing certificate name is clean and professional, not the random original filename.

24.4 Site Manager Acceptance Criteria

- A site manager can log in securely.
- A site manager can see all sites assigned to them.
- A site manager can view certificates across all assigned sites.

- A site manager can filter certificates by one site.
- A site manager sees newest certificates first.
- A site manager can view and download PDFs.
- A site manager cannot see unassigned sites.
- A site manager cannot create users, upload documents, delete documents or edit certificates.

24.5 Security Acceptance Criteria

- A customer cannot access another customer's certificate through the UI.
- A customer cannot access another customer's certificate by guessing or copying a file URL.
- A deactivated user cannot access the portal.
- A customer user cannot access admin routes.
- A site manager cannot access admin routes.
- PDF files are stored in private storage.
- File downloads require permission checks.
- Audit logs are created for certificate upload, view and download events.
- Role-based access is enforced at server/database level, not only in the frontend.

24.6 UX Acceptance Criteria

- Customer users can access the latest certificate quickly without needing training.
- Site managers can switch between assigned sites without logging out.
- Certificate lists default to newest first.
- Download buttons are visible and easy to access.
- Admin upload screens show clear customer/site context.
- Admin users receive confirmation after publishing a certificate.
- The system avoids unnecessary clicks and avoids confusing file names.
- The portal feels like a professional compliance document portal, not a generic file-sharing system.

25. Key Risks and Mitigations

Risk 1 - Customer sees another customer's document

Mitigation: private storage, RLS policies, server-side permission checks, signed URLs and security testing.

Risk 2 - Admin uploads certificate to wrong customer

Mitigation: client-first upload, site-level upload, clear publish confirmation, customer ID display and success confirmation message.

Risk 3 - Portal becomes too complex

Mitigation: no tool-level tracking in MVP, no customer-side user management, no OCR in MVP, keep customer UI simple and phase advanced features later.

Risk 4 - Customers cannot find latest certificate quickly

Mitigation: newest-first default sorting, clear certificate titles, search and filters, visible View/Download buttons and site filters for managers.

Risk 5 - Admin workload remains high

Mitigation: automatic display name generation, client-first upload, site-level upload, recently uploaded view and customer search by ID/name.

26. Future Enhancements

- Bulk certificate upload
- Automated customer email notifications
- Expiry reminder emails
- CSV export for managers
- Advanced reporting
- QR certificate verification integration
- OCR/text extraction from PDFs
- Tool/asset-level tracking
- Engineer upload workflow
- MFA for admins
- Customer support ticket/contact form
- Integration with Oakrange's calibration software

27. Final MVP Statement

The MVP is successful if Oakrange admins can quickly upload certificate PDFs to the correct customer/site, and customers or site managers can securely find, view and download their latest certificates without needing support from Oakrange.

The strongest success test is:

Admin can publish a certificate in under 60 seconds.

Customer can find and download the latest certificate in under 30 seconds.

No customer can access another customer's documents under any circumstances.